

Review of TPAMI-2025-07-2281

1 Summary

The paper presents a systematic literature review (SLR) of model compression techniques (pruning, quantization, distillation, and compact architecture) with an emphasis on sustainability. The authors claim to go beyond prior surveys by systematically analyzing not only compression methods but also supplementary techniques, frameworks, evaluation metrics, parameter tuning, datasets, and models. While the topic is important and the SLR framework is structured, the paper suffers from critical weaknesses: omission of major compression approaches, severe under-coverage, superficial sustainability analysis, and insufficient comparison with existing surveys. For IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), which requires comprehensive, rigorous, and forward-looking surveys, the manuscript does not meet the expected standard.

2 Strengths

1. Topic is timely: compression and sustainability are indeed central in deep learning research.
2. The survey includes 1 statistic (Fig. 3), 3 heatmaps, and more than 40 tables (Tables III-XLVIII) summarizing findings from 39 selected papers.
3. The manuscript follows an SLR methodology, with clear research questions and explicit inclusion/exclusion criteria.

3 Weaknesses and Areas for Improvement

To support my review, I compiled recent surveys on model compression and efficient deep learning in Table 1, including publication year, venue, number of studied cases, number of references (# Refs), and relevant notes. Abbreviations: Transactions on Machine Learning Research (TMLR), IEEE Internet of Things (IoT), IEEE Transactions on Neural Networks and Learning Systems (TNNLS).

Table 1: 33 related surveys of model compression

Survey	Time	Venue	# Cases	# Refs	Areas
A Comprehensive Survey on Knowledge Distillation [15]	Sep 2025	TMLR	N/A	798	Knowledge Distillation

Survey	Time	Venue	# Cases	# Refs	Areas
Vision transformers on the edge: A comprehensive survey of model compression and acceleration strategies [26]	Aug 2025	Neurocomputing	N/A	161	Pruning + Distillation + Quantization for ViT
A review of state-of-the-art techniques for large language model compression [7]	Aug 2025	Complex & Intelligent Systems	N/A	121	Pruning + Distillation + Quantization + NAS for LLM
Stop Overthinking: A Survey on Efficient Reasoning for Large Language Models [29]	Aug 2025	TMLR	N/A	250	Efficient Reasoning for LLM
Vision-Language Models for Edge Networks: A Comprehensive Survey [28]	Aug 2025	IoT Journal	N/A	204	Edge Vision-Language Models
Empowering large language models to edge intelligence: A survey of edge efficient LLMs and techniques [34]	Aug 2025	Computer Science Review	N/A	223	Pruning + Distillation + Quantization + NAS + Decomposition + Compact Architecture for LLM
Tensor decompositions for signal processing: Theory, advances, and applications [32]	Aug 2025	Signal Processing	N/A	252	Tensor Decomposition for CNN, RNN, and Transformers
Efficient compressing and tuning methods for large language models: A systematic literature review [12]	May 2025	ACM Computing Surveys	123	173	Pruning + Distillation + Quantization + Decomposition + PEFT for LLM
On-Device Deep Learning: Survey on Techniques Improving Energy Efficiency of DNNs [1]	May 2025	TNNLS	N/A	497	Pruning + Distillation + Quantization + NAS + Compact Architecture for CNN

Survey	Time	Venue	# Cases	# Refs	Areas
Empowering edge intelligence: A comprehensive survey on on-device ai models [36]	Apr 2025	ACM Computing Surveys	N/A	288	On-Device AI Models
Resource-efficient algorithms and systems of foundation models: A survey [39]	Jan 2025	ACM Computing Surveys	N/A	497	Efficient Foundation Models
Optimizing edge AI: a comprehensive survey on data, model, and system strategies [35]	Jan 2025	Arxiv	N/A	297	Edge AI
A Survey on Deep Neural Network Pruning: Taxonomy, Comparison, Analysis, and Recommendations [4]	Dec 2024	TPAMI	N/A	332	Pruning
A survey on efficient vision transformers: algorithms, techniques, and performance benchmarking [19]	Dec 2024	TPAMI	N/A	85	Pruning + Distillation + Quantization + Compact Architecture for ViT
Review of Lightweight Deep Convolutional Neural Networks [3]	Nov 2024	Archives of Computational Methods in Engineering	N/A	194	Lightweight CNN
A survey of low-bit large language models: Basics, systems, and algorithms [10]	Sep 2024	Neural Networks	N/A	242	LLM Quantization
A comprehensive review of model compression techniques in machine learning [8]	Sep 2024	Applied Intelligence	N/A	293	Pruning + Distillation + Quantization + NAS + Decomposition + Compact Architecture
Green Edge AI: A Contemporary Survey [16]	Jul 2024	Proceedings of the IEEE	N/A	328	Green Edge AI
Lightweight Deep Learning for Resource-Constrained Environments: A Survey [14]	Jun 2024	ACM Computing Surveys	N/A	288	On-Device AI Models

Survey	Time	Venue	# Cases	# Refs	Areas
Efficient Large Language Models: A Survey [33]	May 2024	TMLR	N/A	887	Efficient LLM
Structured Pruning for Deep Convolutional Neural Networks: A Survey [11]	May 2024	TPAMI	N/A	134	Structured pruning CNN
Pruning Deep Neural Networks for Green Energy-Efficient Models: A Survey [31]	May 2024	Cognitive Computation	N/A	179	Pruning CNN
A survey on transformer compression [30]	Apr 2024	Arxiv		289	Pruning + Distillation + Quantization + NAS + Decomposition + Compact Architecture for Transformer
Efficient convolutional neural networks and network compression methods for object detection: a survey [41]	Jan 2024	Multimedia Tools and Applications	N/A	231	CNN compression for object detection
A Comprehensive Survey on Model Quantization for Deep Neural Networks in Image Classification [25]	Dec 2023	ACM Transactions on Intelligent Systems and Technology	N/A	191	CNN Quantization
Efficient deep learning: A survey on making deep learning models smaller, faster, and better [17]	Mar 2023	ACM Computing Surveys	N/A	169	CNN + Transformer compression
Resource-Efficient Convolutional Networks: A Survey on Model-, Arithmetic-, and Implementation-Level Techniques [13]	Feb 2023	ACM Computing Surveys	N/A	198	Pruning + Distillation + Quantization + Compact Architecture + NAS
A review of convolutional neural network architectures and their optimizations [6]	Jan 2023	Artificial Intelligence Review	N/A	311	Lightweight CNNs
Neural architecture search: Insights from 1000 papers [38]	Jan 2023	Arxiv	1000	566	NAS

Survey	Time	Venue	# Cases	# Refs	Areas
Neural Architecture Search and Hardware Accelerator Co-Search: A Survey [27]	Nov 2021	IEEE Access	N/A	178	NAS
Tensor Methods in Computer Vision and Deep Learning [18]	May 2021	Proceedings of the IEEE	N/A	234	Tensor Decomposition
Model Compression and Hardware Acceleration for Neural Networks: A Comprehensive Survey [9]	Apr 2020	Proceedings of the IEEE	N/A	308	Pruning + Distillation + Quantization + Compact Architecture + NAS + Decomposition
A comprehensive survey on model compression and acceleration [5]	Feb 2020	Artificial Intelligence Review		174	Decomposition + Distillation + Quantization + Pruning
This	Sep 2025	Under review	39	90	Distillation + Quantization + Pruning, mostly for CNN

3.1 Scope and Coverage

1. Missing important sub-areas: The survey omits two very important approaches for model compression: low-rank approximation (matrix/tensor decomposition/factorization) [18, 32, 20] and neural architecture search (NAS) [27, 38]. The authors have not explained the reasons for these omissions. Given that RQ1 asks “What types of compression techniques have been applied to DL models?”, the absence of low-rank approximation and NAS undermines the comprehensiveness of the work. The manuscript title, “A Survey of Model Compression Techniques”, is not persuasive if these approaches are missing. The term “model compression” appears 110 times across the manuscript and appendix, but the exclusion of these fundamental methods raises concerns about completeness.
2. The number of studies considered is unconvincing. According to the authors, the search covered publications from January 2017 to February 2025, yielding 1536 papers. After screening, 142 remained, and only 39 were ultimately included in the review. This final number is far too small to reflect the richness of the model compression field. For example, Saha et al. [26] report more than 1000 papers on vision transformer compression in 2024 alone. Similarly, Kim et al. [12], using an SLR approach for large language model (LLM) compression between 2020 and 2024, retained 123 studies for review. For comparison, Table 1 lists 33 surveys published between 2020 and 2025, many of which cover hundreds of references across diverse areas (CNNs, Transformers, LLMs, edge AI, NAS, tensor methods, sustainability). In contrast, the present manuscript restricts itself to just 39 CNN-focused studies and about 90 references, which is insufficient for TPAMI standards.
3. The survey is limited to computer vision, particularly image classification: Subsection IV.J

“Datasets” and Table XIV list datasets almost exclusively from computer vision (image classification, object detection). This omits crucial domains such as NLP (e.g., GLUE, SQuAD), audio/speech processing (LibriSpeech, CommonVoice), and time-series/signal processing benchmarks. A comprehensive survey should include these areas.

4. The survey lacks important neural architectures: Subsection IV.K “Model Architectures” and Table XVI focus mainly on CNNs for image classification and object detection, and some long short-term memory (LSTM) methods (line 19 of page 15). Other major architectures, such as Transformers (for NLP and vision), Graph Neural Networks (GNNs), and Generative Adversarial Networks (GANs), are missing. This is a critical limitation, especially since model compression in LLMs and ViTs is one of the most active areas of current research. Moreover, if we consider only CNN, the survey still misses many important and recent architectures. I cite MobileNetV4: Universal Models for the Mobile Ecosystem [24], published at ECCV 2024, and Yolov10, published at NeurIPS 2024, for example.

3.2 Novelty and Contribution

1. The paper is positioned as “beyond prior surveys,” but in practice, its contribution is modest: it is essentially a categorization exercise with percentages. The related work discussion is limited to only three surveys [17, 37, 2], overlooking many more recent and relevant studies. Since “sustainability” is presented as the distinctive novelty of this work, the comparison should explicitly include surveys such as Boumendil et al. [1] on energy efficiency of DNNs, Wang et al. [36] on on-device AI models, and Mao et al. [16] on green edge AI. Can you provide a full comparison with related surveys (at least 33 surveys in Table 1) in the field? If space limitations in TPAMI prevent including all of them in the main text, such a comparison should at least be provided in the appendix, the code repository, or the response to reviewers.
2. The sustainability discourse is underdeveloped: fewer than a dozen papers explicitly report energy or carbon metrics, making the contribution minimal compared to specialized surveys.
3. Claims of being a “comprehensive SLR” (lines 20-25 of page 2) are exaggerated. Earlier works already provide broad coverage, see Table 1. The claim on the second contribution “A comprehensive overview of current research trends and practices in the field.” is not persuasive given the current coverage of the survey.
4. The stated contribution of offering “valuable guidance for future research” is weak: I identified only four scattered recommendations for future research (line 41 page 7, line 26 page 10, line 51 page 13, line 7 page 18).
5. I checked the code repository on 21 Sep 2025 and found that it contains only PNG and PDF files of figures and tables from the manuscript. The claim that it “supports transparency, reproducibility, and further analysis by the research community” is therefore not substantiated. No code or demonstration is provided to reproduce the research process (e.g., query design, inclusion/exclusion steps). Moreover, recent surveys in top journals typically provide public repositories (e.g., on GitHub) that allow the community to verify and contribute. For example: <https://github.com/hrcheng1066/awesome-pruning> from [4] (TPAMI, 2024); <https://github.com/he-y/Awesome-Pruning> from [11] (TPAMI, 2024, which also offers an interactive demo at <https://huggingface.co/spaces/he-yang/Structured-Pruning-Survey>); and <https://github.com/Eclipsess/Awesome-Efficient-Reasoning-LLMs> from [29] (TMLR, 2025).

3.3 Methodological Issues

1. Search strategy is too shallow: only ACM DL, Scopus, and WoS are used, ignoring IEEE Xplore, arXiv, and Google Scholar, which are primary sources for AI research. This represents a serious limitation in terms of comprehensiveness. Could the authors clarify why IEEE Xplore was excluded from the search databases? For a TPAMI survey, broader coverage is expected to ensure rigor and completeness.
2. Query design (“deep learning” + “compression/optimization”) may miss keywords where terminology differs (e.g., “frugal” for frugal machine learning, “tiny” for tinyML, “lightweight”, “compact”, “mobile”, “edge devices”, which are very common terms in model compression).
3. Why are only 39 works retained, when compression/efficiency is one of the most prolific subfields of deep learning? The drop from 1536 to 39 papers suggests overly aggressive exclusion criteria. This must be justified more rigorously.
4. The “Green AI” discussion is superficial: although environmental sustainability is in the title and abstract, fewer than 12 papers with energy/carbon metrics are reported, and the discussion is minimal. This risks misleading readers about the survey’s true focus.

3.4 Presentation and Analysis

1. Tables are exhaustive but overwhelming, often listing dozens of minor variants with little synthesis. The reader is left with fragmented information rather than clear trends or actionable insights.
2. Figures (heatmaps, distributions) repeat obvious findings (e.g., quantization + pruning dominate). Little interpretation is provided.

4 Minor Issues

1. Table IV lists several hybrid compression strategies, and on page 6, line 59, the authors note “the growing trend toward multi-technique compression pipelines.” I agree with this observation. However, the survey overlooks more recent hybrid approaches for CNN compression, such as [22, 40, 23, 21], which should be included for completeness.
2. Line 51-55: “Although NLP ... from these techniques.”. This claim is inaccurate; model compression is extensively applied in NLP and LLMs. Saying “image classification and object detection now dominate in the application of model compression” is not suitable.
3. Grammar and syntax issues: “Software layers ru on” → “run on”. “Evaluation metrics enables researchers. . .” → should be “enable”. “On inference” (in Table VI, and Table II in appendix) → should be “During inference”, similarly for “On training”. “Time take” in Table XXII → should be “Time taken”. Reference [68]: “Learning multiple layers of features from tiny images” is missing proper formatting for technical report (no italics, inconsistent with others).
4. Line 42 page 16: “This section”. IV.K is a subsection, not a section. Line 50 page 16: “This section”. IV.K.1 is a subsubsection, not a section.

5 Final Recommendation

Major Revision. The manuscript suffers from limited coverage, weak novelty, methodological flaws, and superficial treatment of sustainability, which is supposed to be the central theme. For a journal of TPAMI’s caliber, the bar for comprehensiveness, rigor, and insight is much higher. Although my assessment falls between Reject and Major Revision, I think that this manuscript can be given a chance for a strict and rigorous revision. I recommend a major revision before the paper can be considered for publication.

References

- [1] Anais Boumendil, Walid Bechkit, and Karima Benatchba. “On-Device Deep Learning: Survey on Techniques Improving Energy Efficiency of DNNs”. In: *IEEE Transactions on Neural Networks and Learning Systems* 36.5 (2025), pp. 7806–7821.
- [2] Mingzhuo Cai et al. “Research on compression pruning methods based on deep learning”. In: *Journal of Physics: Conference Series*. Vol. 2580. 1. IOP Publishing. 2023, p. 012060.
- [3] Fanghui Chen et al. “Review of Lightweight Deep Convolutional Neural Networks: F. Chen et al.” In: *Archives of Computational Methods in Engineering* 31.4 (2024), pp. 1915–1937.
- [4] Hongrong Cheng, Miao Zhang, and Javen Qinfeng Shi. “A Survey on Deep Neural Network Pruning: Taxonomy, Comparison, Analysis, and Recommendations”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 46.12 (2024), pp. 10558–10578.
- [5] Tejalal Choudhary et al. “A comprehensive survey on model compression and acceleration”. In: *Artificial Intelligence Review* 53.7 (2020), pp. 5113–5155.
- [6] Shuang Cong and Yang Zhou. “A review of convolutional neural network architectures and their optimizations”. In: *Artificial Intelligence Review* 56.3 (2023), pp. 1905–1969.
- [7] Pierre V Dantas, Lucas C Cordeiro, and Waldir SS Junior. “A review of state-of-the-art techniques for large language model compression”. In: *Complex & Intelligent Systems* 11.9 (2025), pp. 1–40.
- [8] Pierre Vilar Dantas et al. “A comprehensive review of model compression techniques in machine learning”. In: *Applied Intelligence* 54.22 (2024), pp. 11804–11844.
- [9] Lei Deng et al. “Model Compression and Hardware Acceleration for Neural Networks: A Comprehensive Survey”. In: *Proceedings of the IEEE* 108.4 (2020), pp. 485–532.
- [10] Ruihao Gong et al. “A survey of low-bit large language models: Basics, systems, and algorithms”. In: *Neural networks* (2025), p. 107856.
- [11] Yang He and Lingao Xiao. “Structured Pruning for Deep Convolutional Neural Networks: A Survey”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 46.5 (2024), pp. 2900–2919.
- [12] Gun Il Kim, Sunga Hwang, and Beakcheol Jang. “Efficient compressing and tuning methods for large language models: A systematic literature review”. In: *ACM Computing Surveys* 57.10 (2025), pp. 1–39.
- [13] JunKyu Lee et al. “Resource-efficient convolutional networks: A survey on model-, arithmetic-, and implementation-level techniques”. In: *ACM Computing Surveys* 55.13s (2023), pp. 1–36.
- [14] Hou-I Liu et al. “Lightweight deep learning for resource-constrained environments: A survey”. In: *ACM Computing Surveys* 56.10 (2024), pp. 1–42.
- [15] Amir M Mansourian et al. “A Comprehensive Survey on Knowledge Distillation”. In: *Transactions on Machine Learning Research* (2025).

- [16] Yuyi Mao et al. “Green Edge AI: A Contemporary Survey”. In: *Proceedings of the IEEE* 112.7 (2024), pp. 880–911.
- [17] Gaurav Menghani. “Efficient deep learning: A survey on making deep learning models smaller, faster, and better”. In: *ACM Computing Surveys* 55.12 (2023), pp. 1–37.
- [18] Yannis Panagakis et al. “Tensor methods in computer vision and deep learning”. In: *Proceedings of the IEEE* 109.5 (2021), pp. 863–890.
- [19] Lorenzo Papa et al. “A survey on efficient vision transformers: algorithms, techniques, and performance benchmarking”. In: *IEEE transactions on pattern analysis and machine intelligence* 46.12 (2024), pp. 7682–7700.
- [20] Van Tien Pham, Yassine Zniyed, and Thanh Phuong Nguyen. “Coupled Tensor Decomposition for Compact Network Representation”. In: *IEEE Transactions on Neural Networks and Learning Systems* (2025), pp. 1–15.
- [21] Van Tien Pham, Yassine Zniyed, and Thanh Phuong Nguyen. “Efficient tensor decomposition-based filter pruning”. In: *Neural Networks* 178 (2024), p. 106393.
- [22] Van Tien Pham, Yassine Zniyed, and Thanh Phuong Nguyen. “Enhanced Network Compression Through Tensor Decompositions and Pruning”. In: *IEEE Transactions on Neural Networks and Learning Systems* 36.3 (2025), pp. 4358–4370.
- [23] Van Tien Pham, Yassine Zniyed, and Thanh Phuong Nguyen. “Singular values-driven automated filter pruning”. In: *Neural Networks* 192 (2025), p. 107857.
- [24] Danfeng Qin et al. “MobileNetV4: Universal models for the mobile ecosystem”. In: *European Conference on Computer Vision*. Springer. 2024, pp. 78–96.
- [25] Babak Rokh, Ali Azarpeyvand, and Alireza Khanteymoori. “A comprehensive survey on model quantization for deep neural networks in image classification”. In: *ACM Transactions on Intelligent Systems and Technology* 14.6 (2023), pp. 1–50.
- [26] Shaibal Saha and Lanyu Xu. “Vision transformers on the edge: A comprehensive survey of model compression and acceleration strategies”. In: *Neurocomputing* (2025), p. 130417.
- [27] Lukas Sekanina. “Neural architecture search and hardware accelerator co-search: A survey”. In: *IEEE access* 9 (2021), pp. 151337–151362.
- [28] Ahmed Sharshar et al. “Vision-Language Models for Edge Networks: A Comprehensive Survey”. In: *IEEE Internet of Things Journal* 12.16 (2025), pp. 32701–32724.
- [29] Yang Sui et al. “Stop Overthinking: A Survey on Efficient Reasoning for Large Language Models”. In: *Transactions on Machine Learning Research* (2025).
- [30] Yehui Tang et al. “A survey on transformer compression”. In: *arXiv preprint arXiv:2402.05964* (2024).
- [31] Jihene Tmamna et al. “Pruning deep neural networks for green energy-efficient models: A survey”. In: *Cognitive Computation* 16.6 (2024), pp. 2931–2952.
- [32] Neriman Tokcan et al. “Tensor decompositions for signal processing: Theory, advances, and applications”. In: *Signal Processing* 238 (2026), p. 110191.
- [33] Zhongwei Wan et al. “Efficient Large Language Models: A Survey”. In: *Transactions on Machine Learning Research* (2024). Survey Certification.
- [34] Rui Wang et al. “Empowering large language models to edge intelligence: A survey of edge efficient LLMs and techniques”. In: *Computer Science Review* 57 (2025), p. 100755.
- [35] Xubin Wang and Weijia Jia. “Optimizing edge AI: a comprehensive survey on data, model, and system strategies”. In: *arXiv preprint arXiv:2501.03265* (2025).

- [36] Xubin Wang et al. “Empowering Edge Intelligence: A Comprehensive Survey on On-Device AI Models”. In: *ACM Comput. Surv.* 57.9 (Apr. 2025).
- [37] Zeqin Wang et al. “Communication compression techniques in distributed deep learning: A survey”. In: *Journal of Systems Architecture* 142 (2023), p. 102927.
- [38] Colin White et al. “Neural architecture search: Insights from 1000 papers”. In: *arXiv preprint arXiv:2301.08727* (2023).
- [39] Mengwei Xu et al. “Resource-efficient algorithms and systems of foundation models: A survey”. In: *ACM Computing Surveys* 57.5 (2025), pp. 1–39.
- [40] Ziyang Zhang et al. “GCPNet: Gradient-aware channel pruning network with bilateral coupled sampling strategy”. In: *Expert Systems with Applications* 266 (2025), p. 126104.
- [41] Yong Zhou et al. “Efficient convolutional neural networks and network compression methods for object detection: a survey”. In: *Multimedia Tools and Applications* 83.4 (2024), pp. 10167–10209.